### Photon-number qubits

While it is natural to encode single qubits into single photons, this is not the only option for realising qubits with light. In principle, any two orthogonal states will do as a basis. The perhaps simplest choice for embedding a qubit in a harmonic oscillator (e.g. light) is to pick the two lowest energy eigenstates, i.e. the Fock states $|0\rangle$ and $|1\rangle$. These correspond to the absence or presence of a single photon in a given mode. In this basis, $z$-rotations (generated by $\hat{\sigma}_z$) are easy to implement as the correspond to simple phase shifts. However, $x$-rotations (generated by $\hat{\sigma}_x$) are hard because they require changing the photon number (hence the energy). Similarly, measurements of $\hat{\sigma}_z$ correspond simply to single-photon detection, while measurements of $\hat{\sigma}_x$ (or in other rotated bases) are challenging.

### Coherent-state qubits and cats

Coherent states are some of the easiest states to generate in practice, since they correspond to the output of lasers. Unfortunately, two coherent states are never perfectly orthogonal (as seen from (3.73)) so they cannot be used to define a perfect qubit. However, for large amplitudes $|\alpha\rangle$ and $|-\alpha\rangle$ are almost orthogonal (the overlap decreases exponentially with $|\alpha|^2$) and they can then be used as good approximations for the computational qubit states $|0\rangle$, $|1\rangle$. In this encoding, $\hat{\sigma}_z$ gates are easy to implement since they just correspond to phase shifts by $\pi$, and measurements of $\hat{\sigma}_z$ are also simple. However, $\hat{\sigma}_x$ is tricky since it must transform between the coherent states $|\pm\alpha\rangle$ and the cat states (3.123), which cannot be done with linear optics. Nevertheless, it has been shown that combining linear optics with photon counting and additional cat states as a resource, universal quantum computing can in principle be achieved.

With current technology, quantum computing with coherent and cat states is (still) not practical in the optical regime. However, in the microwave regime, using superconducting circuits, approaches based on such states are being actively pursued.

### GKP qubits

Another way to encode qubits into the continuous degrees of freedom of the quantum harmonic oscillator (in particular light or microwaves) is the Gottesman-Kitaev-Preskill (GKP) code, first proposed by Daniel Gottesman, Alexei Kitaev, and John Preskill in 2001. The GKP encoding uses a single optical mode and defines logical qubit states as superpositions of peaks in phase space. This provides robustness again loss and noise and enables error correction via linear optical operations. In ideal GKP states, there are infinitely many peaks and each peak is infinitely squeezed in both quadratures. Hence, such states are not physical. However, they can be approximated by states of finite squeezing and energy, which still retain good error-correction properties. GKP states are central to continuous-variable quantum computing, and we will return to them in Chap. 7.

# Preface

Quantum information science and technology is a rapidly developing field that has seen tremendous progress and growth in recent years. The field ranges from fundamental research to applied technology that is now being commercialised or, in some cases, is already available on the market.

These notes have been developed for the masters-level course *Quantum Information Technology* at the Technical University of Denmark (DTU) and is part of the core curriculum in the Master in Quantum Information Science offered jointly by the DTU and the University of Copenhagen. This course builds on top of introductory courses in quantum information and quantum computing and covers advanced topics not usually included in a first quantum information course. The focus is on applications in sensing, computing, and secure communication. Readers are thus assumed to be familiar with quantum mechanics and with basic ideas from quantum information and computing. An introductory chapter is included, covering the main elements of quantum theory required throughout the notes. This is intended mostly as a recap and to establish notation – but motivated readers less familiar with the basics might also be able to start directly from there.

Quantum information technology is a vast area as both the range of applications of quantum information protocols and the range of different physical platforms used to implement them is wide. These notes do not pretend to attempt a complete overview of the entire field. Instead, the focus is on fundamental concepts and a handful of important technological application mostly centered on optical platforms. In particular, quantum optical interferometry and colour-centre magnetometry for sensing, measurement-based and continuous-variable optical quantum computing, and device-independent and continuous-variable optical quantum randomness certification and key distribution. Some important topics *not* covered here include superconducting platforms and ion-trap quantum computing, quantum dots, quantum sensing with atomic ensembles, and atomic clocks.

The quantum-information-technology course is still rather new – it ran for the first time in its current form in 2024 – and these notes are a work in progress that will continuously be updated, expanded, and improved upon. Any feedback and information about errors or typos is highly appreciated and can be addressed to either of the authors.

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